Problem Set 2

Instructor solutions

Counting, events, and shared birthdays

100 points + 10 optional challenge points

Teaching note. This assignment revisits the probability spaces from week two and connects them to counting and inclusion–exclusion. Allow approximately three hours, excluding the challenge. Subpart scores appear in parentheses. Give credit for any correct model and reasoning; distinguish a wrong model from an arithmetic error carried through correctly.

Instructions. Identify the outcomes, their probabilities, and the event being counted. State whether order matters and repetition is allowed. Give exact answers except where decimal approximations are requested. A calculator or short program may be used for Problem 7. Show enough work that the model and the calculation can be checked.

I. Probability spaces: outcomes need weights

1. A weighted finite space

[12 points]

Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, let $\mathcal{F} = 2^{\Omega}$ be all subsets, and suppose $\mathbb{P}(\{i\}) = ci$ for $i = 1, \dots, 6$.

(a) Find c. Explain why these masses define a probability space when $\mathbb{P}(E)$ is the sum of the masses in E. (4)

(b) Let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. Find $\mathbb{P}(A)$, $\mathbb{P}(B)$, $\mathbb{P}(A \cap B)$, and $\mathbb{P}(A \cup B)$. Check the last answer by inclusion–exclusion. (4)

(c) A student says $\mathbb{P}(A) = |A|/|\Omega| = 1/2$. Explain the error and state the extra assumption that would make this counting rule valid. (4)

Solution

(a) The total mass is $c(1 + 2 + 3 + 4 + 5 + 6) = 21c$, so $c = 1/21$. All masses are nonnegative and sum to one. Summing them over disjoint events is additive because no outcome is counted twice. On a finite space only finitely many members of a disjoint family are nonempty, so this also verifies countable additivity. The collection of all subsets is closed under complements and countable unions.

(b) Summing weights gives

$$\mathbb{P}(A) = \frac{12}{21} = \frac{4}{7}, \quad \mathbb{P}(B) = \frac{15}{21} = \frac{5}{7}, \quad \mathbb{P}(A \cap B) = \frac{10}{21}, \quad \mathbb{P}(A \cup B) = \frac{17}{21}.$$

The overlap check is $(12 + 15 - 10)/21 = 17/21$.

(c) The six outcomes are not equally likely: outcome i has mass $i/21$. A set's probability is the sum of its masses, not its relative cardinality. The rule $\mathbb{P}(A) = |A|/|\Omega|$ is valid when the model assigns equal probability to every individual outcome.

2. The same recorded outcomes, two experiments

[10 points]

Experiment I flips two fair coins independently and records the number K of heads. Experiment II chooses an integer uniformly from $\{0, 1, 2\}$.

(a) Give the ordered sample space for Experiment I and the probability of each ordered outcome. Then give the probabilities of the recorded values $K = 0, 1, 2$. (4)

(b) Find the probability of recording at least one in each experiment. Why does the common recorded sample space $\{0, 1, 2\}$ not force the two answers to agree? (3)

(c) After Experiment II records k, choose uniformly among the two-coin strings with exactly k heads. Find the probabilities of HH, HT, TH, TT. Does this reproduce Experiment I? (3)

Solution

(a) The ordered space is $\{HH, HT, TH, TT\}$, with all subsets as events and mass $1/4$ per outcome. Recording K gives probabilities $1/4, 1/2, 1/4$ on $0, 1, 2$, respectively: two strings yield one head.

(b) Experiment I gives $3/4$, while Experiment II gives $2/3$. A sample space specifies possible outcomes, not their weights.

(c) Experiment II assigns probability $1/3$ to $k = 0$ and $k = 2$, producing TT and HH, respectively. The probability $1/3$ of $k = 1$ is split equally between HT and TH. Thus the four probabilities in the requested order are $1/3, 1/6, 1/6, 1/3$. They are not all $1/4$, so the resulting experiment is different.

II. Permutations and binomial coefficients

3. What distinguishes two selections? [12 points]

Explain the counting rule in each part.

(a) Choose a three-person committee from eight students. (3)

(b) Assign president, secretary, and treasurer from those eight students, with no person holding two offices. (3)

(c) Choose a three-person committee from those students and designate one member as chair; the other two have no distinct roles. (3)

(d) Arrange the seven letters of `BALLOON`, treating repeated copies of the same letter as indistinguishable. (3)

Solution

(a) Order is ignored and repetition is forbidden, so the count is $\binom{8}{3} = 56$.

(b) The offices distinguish the three positions. There are $8 \cdot 7 \cdot 6 = 336$ ordered selections without repetition.

(c) Choose the committee and then its chair: $3\binom{8}{3} = 168$. Equivalently, choose the chair first and the other two members next: $8\binom{7}{2} = 168$.

(d) There are two L's and two O's. Labeling them temporarily gives 7! arrangements, but each unlabeled arrangement occurs 2!2! times. The required count is $7!/(2!2!) = 1260$.

4. Why binomial coefficients appear [12 points]

(a) Prove Pascal's identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ for $1 \le k \le n-1$ by a counting argument, not by manipulating factorials. (4)

(b) Prove $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ by counting all subsets of an n-element set in two ways. (4)

(c) Find the coefficient of x^3 in $(1 + 2x)^6$. Explain both the binomial coefficient and the power of 2 in your answer. (4)

Solution

(a) Distinguish one of the n objects. Subsets of size k that contain it are obtained by choosing $k-1$ of the other $n-1$ objects. Subsets that omit it are obtained by choosing k of those $n-1$ objects. The two disjoint cases exhaust all k-element subsets, proving the identity.

Solution (continued)

(b) Classifying subsets by size gives the sum on the left. Independently choosing whether to include each of the n objects gives 2^n subsets, including the empty set. Both expressions count the same set.

(c) Choose which three of the six factors contribute $2x$; the other factors contribute 1. Each such choice contributes 2^3x^3. Hence the coefficient is $\binom{6}{3}2^3 = 20 \cdot 8 = 160$.

III. Inclusion–exclusion: cardinalities and probabilities

5. Overlaps among divisibility conditions [12 points]

Let A, B, C be the integers from 1 through 300 divisible by $4, 6, 10$, respectively. Here "through 300" includes both endpoints.

(a) Find the sizes of the three sets, their three pairwise intersections, and their triple intersection. (4)

(b) How many integers are divisible by at least one of $4, 6, 10$? How many are divisible by none? (4)

(c) How many are divisible by exactly two of $4, 6, 10$? Explain how you remove the triple intersection from the pairwise counts. (4)

Solution

(a) Divisibility by both r and s means divisibility by their lcm.

Set	A	B	C	$A \cap B$	$A \cap C$	$B \cap C$	$A \cap B \cap C$
Required divisor	4	6	10	12	20	30	60
Count	75	50	30	25	15	10	5

Each count is $\lfloor 300/d \rfloor$ for the indicated divisor d.

(b) The union has $75 + 50 + 30 - 25 - 15 - 10 + 5 = 110$ elements. Its complement has $300 - 110 = 190$.

(c) The pairwise counts total $25 + 15 + 10 = 50$. An element in exactly two sets contributes once; one in all three contributes three times. To keep only exactly-two elements, subtract $3 \cdot 5$, giving $50 - 15 = 35$.

6. Can these event probabilities occur? [14 points]

Three events have the following probabilities; no independence is assumed.

Event	A	B	C	$A \cap B$	$A \cap C$	$B \cap C$	$A \cap B \cap C$
Probability	10/20	9/20	8/20	4/20	3/20	2/20	1/20

(a) Find the probability of at least one event and the probability of none. (4)

(b) Find the probability of exactly one event. (4)

(c) Verify that the data define a possible model by giving nonnegative probability masses for the eight disjoint membership regions and checking that they sum to one. (6)

Solution

(a) Inclusion–exclusion gives $\mathbb{P}(A \cup B \cup C) = (10 + 9 + 8 - 4 - 3 - 2 + 1)/20 = 19/20$. The probability of none is $1/20$.

(b) The A-only mass is $(10 - 4 - 3 + 1)/20 = 4/20$. Similarly the B-only and C-only masses are each $4/20$. These regions are disjoint, so the exactly-one probability is $12/20 = 3/5$.

Solution (continued)

(c) The region masses are

Region	Mass	Region	Mass
A only	4/20	A and B only	3/20
B only	4/20	A and C only	2/20
C only	4/20	B and C only	1/20
All three	1/20	None	1/20

They are nonnegative and sum to 20/20. Take these eight regions as the eight outcomes of a finite probability space, with all subsets measurable and these masses. Define A, B, C by the corresponding membership labels. Their marginal and intersection probabilities are exactly the supplied values, so the data are realizable.

IV. Birthday collisions

7. A classroom birthday model [16 points]

Assume n labeled people have independent birthdays, each uniform over 365 days. Ignore leap days.

(a) Specify $\Omega, \mathcal{F}$, and the probability of each individual outcome. For $1 \le n \le 365$, count the outcomes with all birthdays distinct. (4)

(b) Derive an exact formula for the probability $q(n)$ of at least one shared birthday. What is the answer when $n > 365$? (4)

(c) Compute $q(22)$ and $q(23)$ to four decimal places. Explain why these values establish the smallest class size for which a collision is more likely than not. (4)

(d) For $n = 23$, compute the probability that someone else shares one designated member's birthday. Explain why it is much smaller than $q(23)$. (4)

Solution

(a) $\Omega = \{1, \dots, 365\}^n$, $\mathcal{F} = 2^\Omega$, and each ordered list has probability 365^{-n} by the independent uniform model. For all birthdays to be distinct, there are $365 \cdot 364 \cdots (365 - n + 1) = (365)_n$ lists.

(b) Divide the no-collision count by 365^n and take the complement:

$$q(n) = 1 - \frac{(365)_n}{365^n} = 1 - \prod_{j=0}^{n-1} \left(1 - \frac{j}{365}\right) \quad (1 \le n \le 365).$$

For $n > 365$, at least two people must share a day, so $q(n) = 1$.

(c) $q(22) \approx 0.4757$ and $q(23) \approx 0.5073$. Adding a person cannot remove an existing match, so $q(n)$ is nondecreasing. Consequently no class with at most 22 people has probability above 1/2, while a class of 23 does.

(d) Conditional on the designated person's day, each of the 22 others avoids it with probability 364/365, independently. This conditional probability is the same for every possible designated day, so the unconditional match probability is also

$$1 - (364/365)^{22} \approx 0.0586.$$

There are just 22 comparisons with that person, whereas a match anywhere in the group can involve any of $\binom{23}{2} = 253$ pairs. The pair events overlap, so the larger number of pairs explains the contrast without implying that their probabilities may simply be added.

8. Three people: inclusion–exclusion checks the product [12 points]

In a toy calendar of ten equally likely days, three labeled people have independent birthdays. Let E_{ij} be the event that people i and j share a birthday.

(a) Find $\mathbb{P}(E_{12})$, the probability of each pairwise intersection among E_{12}, E_{13}, E_{23}, and the probability of their triple intersection. Explain the event described by an intersection. (5)

(b) Use inclusion–exclusion to find the probability of at least one match. Check it by counting the complement. (4)

(c) A student computes the no-match probability as $(9/10)^3$ by multiplying the three pairwise no-match probabilities. Explain why that multiplication is not justified. (3)

Solution

(a) A single pair matches with probability $1/10$. Any two pair-match events force all three birthdays to agree; the triple intersection describes that same event. There are 10 all-equal lists among 10^3 equally likely lists, so every pairwise intersection and the triple intersection have probability $10/10^3 = 1/100$.

(b) Inclusion–exclusion gives

$$3 \cdot \frac{1}{10} - 3 \cdot \frac{1}{100} + \frac{1}{100} = \frac{7}{25} = .28.$$

The complement check is $1 - (10 \cdot 9 \cdot 8)/10^3 = 1 - .72 = .28$.

(c) Multiplication would require mutual independence of the no-match events. They are not mutually independent: their joint probability is .72, while the proposed product is .729. Structurally, after people 1 and 2 have different birthdays, avoiding both requires person 3 to avoid two distinct days. Individual birthdays are independent, but these overlapping comparisons are not.

Optional challenge: nobody gets their own hat [10 extra points]

The n labeled hats of n people are returned as a uniformly random permutation. Use inclusion–exclusion to show that the probability that no one gets their own hat is

$$\sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Explain the count for an intersection of k specified correct-hat events, and evaluate the probability when $n = 4$.

Solution

Let A_i mean that person i receives their own hat. An intersection of k specified events fixes those k hats; the other $n - k$ hats may be permuted in $(n - k)!$ ways. There are $\binom{n}{k}$ ways to specify the fixed hats. Inclusion–exclusion for the complement of the union therefore gives the no-fixed-point count

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Divide by the $n!$ equally likely permutations. At $n = 4$, $1 - 1 + 1/2 - 1/6 + 1/24 = 3/8$; equivalently, 9 of the 24 permutations have no correct hats.
Suggested credit: intersection counts (4), inclusion–exclusion and normalization (4), evaluation at $n = 4$ (2).